Module 13: Week 15

1. Self-driving cars like Tesla Autopilot (Uses AI)

Self-driving cars are a complex real-world application of AI. They must perceive their environment, make decisions, and control the vehicle in real time.

- **Perception**: AI uses computer vision to interpret images from cameras, detect objects like pedestrians, road signs, and other cars. Supervised learning is used where models are trained on labeled images.
- **Decision-Making**: The car needs to decide when to change lanes, yield, or stop based on the current environment. This is often done using reinforcement learning, where the system learns optimal actions through trial and error in simulated environments.
- **Control**: For tasks like steering, accelerating, or braking, control systems use AI models that adjust based on learned driving policies and sensor feedback.

Conclusion: AI is used heavily in all components of self-driving cars. It combines Supervised Learning and Reinforcement Learning.

2. Mobile banking app displaying your current account balance (Does not use AI)

The mobile banking app simply retrieves and displays your account data from a central banking server or database. There is no decision-making, prediction, pattern recognition, or learning involved.

It does not analyze your behavior or adapt based on your past activity.

No algorithms are used to classify or cluster data, nor does it try to predict future transactions or provide recommendations.

Conclusion: This is a deterministic application where it performs predefined tasks without any learning. Therefore, no AI or ML is involved.

3. Netflix recommending movies and TV shows based on your viewing history (Uses AI)

Netflix uses AI-based recommendation systems to personalize content suggestions for each user. It studies your viewing habits, watch time, genres preferred, and how similar users behave.

Supervised Learning is used when the system is trained on labeled data, for example, user ratings or likes/dislikes.

Unsupervised Learning may be used to find hidden patterns or group similar users (user clustering).

In some cases, Reinforcement Learning can be used to optimize recommendations based on real-time user interaction i.e. what users click on after a recommendation.

Conclusion: AI is used for Netflix's user experience. It uses Supervised and Unsupervised Learning to predict preferences and personalize recommendations.

4. Barcode scanner at a supermarket checkout (Does not use AI)

A barcode scanner simply reads the black and white pattern on a barcode using a light sensor and matches it to a product in a predefined database.

It does not adapt or learn from usage or history.

It always returns the same product information for the same barcode.

Conclusion: The process is purely rule-based and deterministic, not intelligent. There is no AI or ML involved.

5. YouTube’s content moderation system that flags potentially harmful videos (Uses AI)

YouTube uses AI models to detect inappropriate content, such as hate speech, violence, or misinformation.

These systems are trained using Supervised Learning, where large datasets of labeled videos (e.g., "harmful", "not harmful") are used to teach the model what to flag.

The AI analyzes audio (speech-to-text), visual content (image/video recognition), and metadata to make judgments.

Sometimes, Reinforcement Learning may be used to refine moderation decisions based on feedback from human reviewers.

Conclusion: Youtube is a clear example of AI, particularly Supervised Learning, because the system learns from examples to make future predictions.